

Assignment 5

Ethan

Set Up

Packages

```
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

#Wrap Axis Label
library(scales)

#reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# visualizing
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(
```

```

    here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
    sheet = "Aquatic Insects"
  )

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)

```

Necessary Code from Class

```

# creating new object from sites
ncos_sites <- sites %>%
  # filtering to only include sites at NCOS, sampled quarterly
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# creating new object from taxon_list
taxon_list_clean <- taxon_list %>%
  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))

water_quality_clean <- water_quality %>%
  clean_names() %>%
  semi_join(ncos_sites, by = "site") %>%
  unite("date_site", date, site, remove = TRUE) %>%
  group_by(date_site) %>%

```

```

summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) %>%
ungroup() %>%
filter(date_site != "2022-08-12NVBR")

aq_ins_clean <- aq_ins %>%
  clean_names() %>%
  semi_join(ncos_sites, by = "site") %>%
  rename(date = date_on_vial) %>%
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         annelida_polychaete,
         diptera_chironomid) %>%
  filter(sample_type %in% c("FB250", "FB 250")) %>%
  pivot_longer(
    # columns to make longer
    cols = ostracod:annelida_polychaete,
    # new column for the old column names
    names_to = "taxon",
    # new column for the values in each of the old columns
    values_to = "abundance"
  ) %>%
  group_by(date, site, taxon) %>%
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) %>%
  ungroup() %>%
  unite("date_site", date, site, remove = FALSE) %>%
  left_join(water_quality_clean, by = "date_site") %>%
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) %>%
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "East Channel",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",

```

```

    site == "NDC" ~ "Devereux Creek")) %>%
mutate(site_full = fct_reorder(site_full,
                                med_sal,
                                .fun = "median",
                                .na_rm = TRUE),
       site_full = fct_rev(site_full))

copepods <- aq_ins_clean %>%
  filter(taxon == "copepod") %>%
  mutate(log_ave_lit = log(ave_lit + 1))

```

1. Visualizing differences across sites in major taxon

a. Planning the Figure

- x-axis: Salinity (ppt)
- y-axis: taxon
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

b. Wrangling the data

```

ostracods <- aq_ins_clean %>%
  filter(taxon == "ostracod") %>%
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods, n = 10)

```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	2023-01-28_NPB1	2023-01-28 00:00:00	NPB1	ostr~	0	7.66	NA	9.74
2	2022-05-13_NVBR	2022-05-13 00:00:00	NVBR	ostr~	0	9.72	74.2	11.4
3	2020-01-21_NDC	2020-01-21 00:00:00	NDC	ostr~	2.4	6.53	0.3	7.6
4	2021-09-01_NPB2	2021-09-01 00:00:00	NPB2	ostr~	0	NA	NA	NA
5	2021-11-23_NPB2	2021-11-23 00:00:00	NPB2	ostr~	0.267	7.59	1.99	0.65
6	2023-03-03_NDC	2023-03-03 00:00:00	NDC	ostr~	5.73	7.45	0.688	5.64

```

7 2022-11-19_NDC 2022-11-19 00:00:00 NDC ostr~ 6 NA NA NA
8 2022-01-24_NEC 2022-01-24 00:00:00 NEC ostr~ 0.267 NA NA NA
9 2022-08-15_NMC 2022-08-15 00:00:00 NMC ostr~ 0.133 7.61 NA NA
10 2022-11-18_NVBR 2022-11-18 00:00:00 NVBR ostr~ 0 NA NA NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
# Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
# Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
# comments <chr>, site_full <fct>, log_ave_lit <dbl>

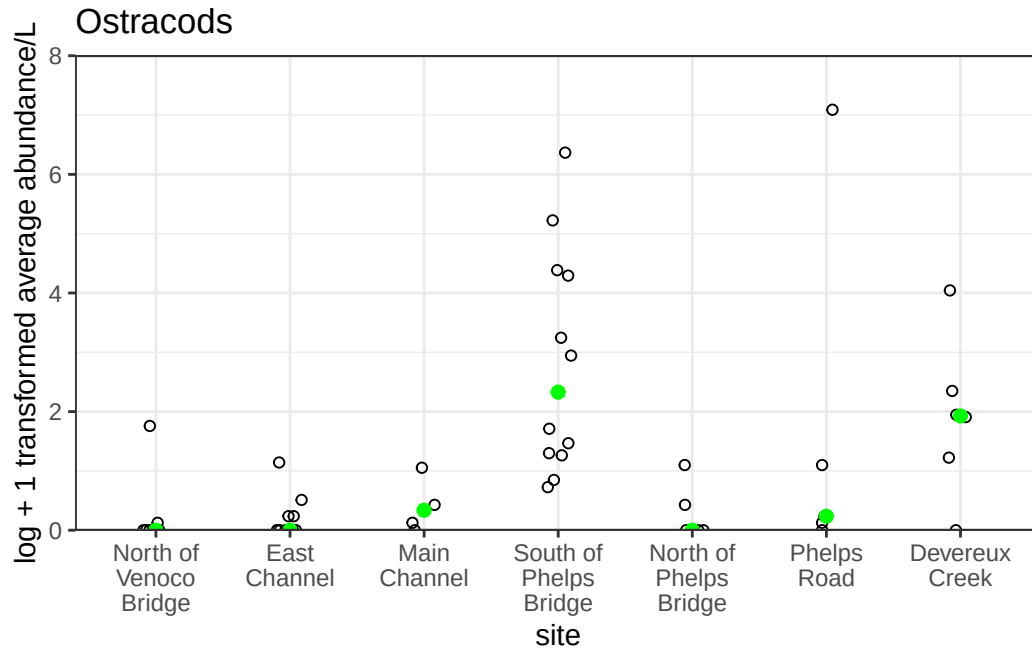
```

c. Coding the figure

```

ostracod_plot <- ggplot(data = ostracods,
                        mapping = aes(x = site_full,
                                      y = log_ave_lit))+
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  stat_summary(geom = "point",
              fun = median,
              color = "green",
              size = 2)+
  scale_x_discrete(labels = label_wrap(width = 10)) +
  scale_y_continuous(expand = c(0, 0),
                    limits = c(0, 8)) +
  labs(x = "site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  theme_bw()
ostracod_plot

```



d. Creating a multipanel figure

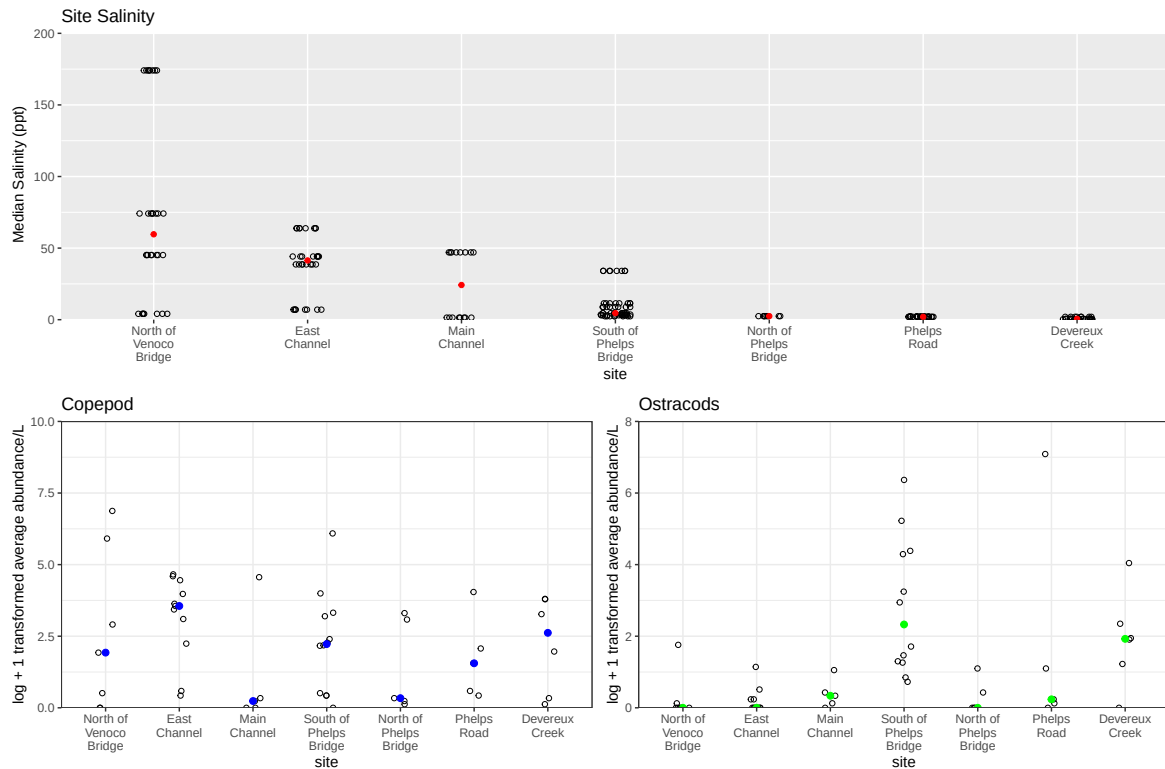
```
# creating an object of a salinity plot
salinity_plot <- ggplot(data = aq_ins_clean,
                        mapping = aes(x = site_full,
                                      y = med_sal)) +

  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  stat_summary(geom = "point",
              fun = median,
              color = 'red') +
  scale_x_discrete(labels = label_wrap(width = 10)) +
  scale_y_continuous(expand = c(0, 0),
                    limits = c(0, 200)) +
  labs(x = "site",
       y = " Median Salinity (ppt)",
       title = "Site Salinity")
```

```
# creating a plotting object of copepod abundance
copepod_plot <- ggplot(data = copepods,
                      mapping = aes(x = site_full,
                                    y = log_ave_lit))+

  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  stat_summary(geom = "point",
              fun = median,
              color = "blue",
              size = 2)+
  scale_x_discrete(labels = label_wrap(width = 10)) +
  scale_y_continuous(expand = c(0, 0),
                    limits = c(0, 10)) +
  labs(x = "site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepod") +
  theme_bw()
```

```
salinity_plot /
(copepod_plot | ostracod_plot)
```



e. Interpreting the figure

The biggest differences in copepod and ostracod abundance across sites can be seen at the East Channel site, where copepods are highest in abundance, and ostracods are extremely low in abundance. The two species are similar in abundance at the Main Channel, South of Phelps Bridge, North of Phelps Bridge, and Devereux Creek sites.

There doesn't appear to be a correlation between site salinity and copepod or ostracod abundance. Median salinity is order by descending ppt from left to right on the x-axis, and the invertebrate abundance plots with the same x-axis don't follow a descending or ascending pattern, appearing relatively random.

2. Visualizing total abundance as a function of salinity

a. Planning the Figure

- x-axis: Salinity (ppt)
- y-axis: log_avg_lit

- color: taxon
- geometry: geom_points()
- facets: facet_wrap(~ taxon)

b. Wrangling the Data

```
avg_taxon_abundance <- aq_ins_clean %>%
  select(date_site, taxon, med_sal, ave_lit) %>%
  mutate(
    taxon = to_any_case(taxon, case = "title"),
    log_ave_lit = log(ave_lit + 1),
    taxon = fct_reorder(taxon, ave_lit, .fun = "max", .na_rm = TRUE)
  )

slice_sample(avg_taxon_abundance, n = 10)
```

A tibble: 10 x 5

	date_site	taxon	med_sal	ave_lit	log_ave_lit
	<chr>	<fct>	<dbl>	<dbl>	<dbl>
1	2023-08-12_NEC	Annelida Polychaete	NA	0	0
2	2023-05-04_NMC	Annelida Polychaete	NA	0	0
3	2023-03-03_NPB2	Copepod	1.98	0.533	0.427
4	2023-11-11_NEC	Nematode	38.6	0	0
5	2022-05-12_NPB	Diptera Ceratopogonidae	11.4	0	0
6	2023-03-03_NPB2	Ostracod	1.98	0.133	0.125
7	2023-11-11_NEC	Diptera Ceratopogonidae	38.6	0.133	0.125
8	2023-03-03_NPB2	Hemiptera Corixidae Boatman	1.98	0	0
9	2023-07-17_NPB	Cladocera	NA	0	0
10	2023-05-06_NDC	Diptera Ceratopogonidae	NA	0	0

c. Coding the figure